1

Renal Anatomy

Wilhelm Kriz, Marlies Elger

The complex structure of the mammalian kidney is best understood in the unipapillary form that is common to all small mammalian species. Fig. 1.1 is a schematic coronal section through a unipapillary kidney, with a cortex enclosing a pyramid-shaped medulla, the tip (papilla) of which protrudes into the renal pelvis. The medulla is divided into an outer and an inner medulla; the outer medulla is further subdivided into an outer and an inner stripe.

STRUCTURE OF THE KIDNEY

The specific components of the kidney are the nephrons, the collecting ducts, and a unique microvasculature.[1] The multipapillary kidney of humans contains about 1 million nephrons, although this number varies considerably. The number of nephrons is established during prenatal development; after birth, new nephrons cannot be developed, and lost nephrons cannot be replaced.

Nephrons

A nephron consists of a renal corpuscle *(glomerulus)* connected to a complicated and twisted tubule that finally drains into a collecting duct (Fig. 1.2 and Table 1.1). Three types of nephrons can be distinguished by the location of renal corpuscles within the cortex: superficial, midcortical, and juxtamedullary nephrons. The tubular part of the nephron consists of a proximal tubule and a distal tubule connected by a loop of Henle (for details see later, as well as Kriz and Bankir[2]). Superficial and midcortical nephrons have short loops that turn back in the outer medulla or even in the cortex. Juxtamedullary nephrons have long loops that turn back at successive levels of the inner medulla.

Collecting Ducts

A collecting duct is formed in the renal cortex when several nephrons join. A connecting tubule is interposed between a nephron and a cortical collecting duct. Cortical collecting ducts descend within the medullary rays of the cortex. They then traverse the outer medulla as unbranched tubes. On entering the inner medulla, they fuse successively and open finally as papillary ducts into the renal pelvis (see Fig. 1.2 and Table 1.1).

Microvasculature

The microvascular pattern of the kidney is similarly organized in mammalian species[1,3] (Fig. 1.3; see also Fig. 1.1). The renal artery, after entering the renal sinus, finally divides into the interlobar arteries, which extend toward the cortex in the space between the wall of the pelvis (or calyx) and the adjacent cortical tissue. At the junction between cortex and medulla, the interlobar arteries divide and pass over into the arcuate arteries, which also branch. The arcuate arteries give rise to the cortical radial arteries (interlobular arteries), which ascend radially through the cortex. No arteries penetrate the medulla.

Afferent arterioles supply the glomerular tufts and generally arise from cortical radial arteries. *Aglomerular* tributaries to the capillary plexus are rarely found. As a result, the blood supply of the peritubular capillaries of the cortex and the medulla is exclusively *postglomerular.*

Glomeruli are drained by efferent arterioles. Two basic types of efferent arterioles can be distinguished: *cortical* efferent arterioles, which derive from superficial and midcortical glomeruli, supply the capillary plexus of the cortex. Efferent arterioles of *juxtamedullary* glomeruli supply the renal medulla. Within the outer stripe of the medulla, these vessels divide into the *descending* vasa recta and then penetrate the inner stripe in cone-shaped vascular bundles. At intervals, individual vessels leave the bundles to supply the capillary plexus at the adjacent medullary level. A fraction of the descending vasa recta enters the inner medulla; few of them reach the papillary region.

Ascending vasa recta drain the renal medulla. In the inner medulla, the vasa recta arise at every level, ascending as unbranched vessels, and traverse the inner stripe within the vascular bundles. The ascending vasa recta that drain the inner stripe may join the vascular bundles; the majority ascends directly to the outer stripe between the bundles. All the ascending vasa recta traverse the outer stripe as individual wavy vessels with wide lumens interspersed among the tubules. Because true capillaries derived from direct branches of efferent arterioles are relatively scarce, the ascending vasa recta form the capillary plexus of the outer stripe. The ascending vasa recta empty into arcuate veins.

The vascular bundles represent a countercurrent exchanger between the blood entering and leaving the medulla. The organization of the vascular bundles also separates the blood flow to the inner stripe from the blood flow to the inner medulla. Descending vasa recta supplying the inner medulla traverse the inner stripe within the vascular bundles. Therefore blood flowing to the inner medulla has not been exposed previously to tubules of the inner or outer stripe. All ascending vasa recta originating from the inner medulla traverse the inner stripe within the vascular bundles. Thus blood that has perfused tubules of the inner medulla does not subsequently perfuse tubules of the inner stripe. However, the blood returning from either the inner medulla or the inner stripe afterward does perfuse the tubules of the outer stripe. This arrangement in the outer stripe may function as the ultimate trap to prevent solute loss from the medulla.

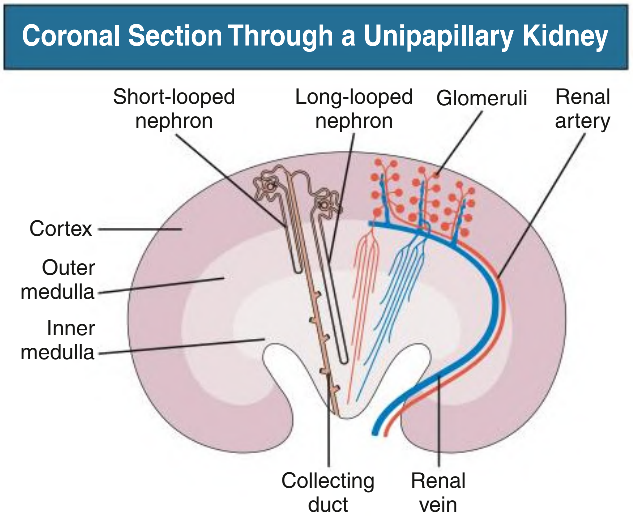

Fig. 1.1 Coronal section through a unipapillary kidney.

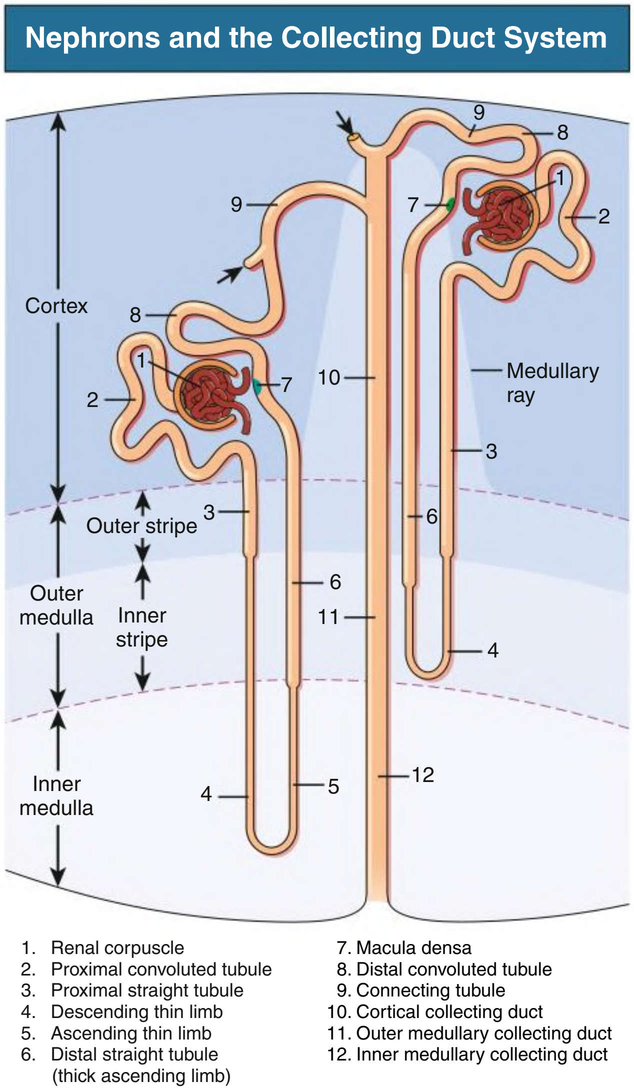

Fig. 1.2 Nephrons and the Collecting Duct System. Shown are short-looped and long-looped nephrons, together with a collecting duct (not drawn to scale). *Arrows* denote confluence of further nephrons.

TABLE 1.1 Subdivisions of the Nephron and Collecting Duct System

Section	Subsections
Nephron	
Renal corpuscle	*Glomerulus:* term used most frequently to refer to entire renal corpuscle Bowman's capsule
Proximal tubule	Convoluted part Straight part (pars recta) or thick descending limb of Henle loop
Intermediate tubule	Descending part or thin descending limb of Henle loop Ascending part or thin ascending limb of Henle loop
Distal tubule	Straight part or thick ascending limb of Henle loop: subdivided into medullary and cortical parts; the cortical part contains the macula densa in its terminal portion Convoluted part
Collecting Duct System	
Connecting tubule	Includes the arcades in most species
Collecting duct	Cortical collecting duct Outer medullary collecting duct: subdivided into an outer stripe and an inner stripe portion Inner medullary collecting duct: subdivided into basal, middle, and papillary portions

Central to the renal drainage of the kidney are the arcuate veins, which, in contrast to arcuate arteries, do form real anastomosing arches at the corticomedullary border. The arcuate veins accept the veins from the cortex and the renal medulla. The arcuate veins join to form interlobar veins, which run alongside the corresponding arteries.

The intrarenal arteries and the afferent and efferent glomerular arterioles are accompanied by sympathetic nerve fibers and terminal axons representing the efferent nerves of the kidney.[1] Tubules have direct contact to terminal axons only when the tubules are located around the arteries or the arterioles. Tubular innervation consists of occasional fibers adjacent to perivascular tubules.[4] The density of nerve contacts to convoluted proximal tubules is low; contacts to straight proximal tubules, thick ascending limbs of Henle loops, and collecting ducts (located in medullary rays and outer medulla) have never been encountered. The vast majority of tubular portions have no direct relationships to nerve terminals. Afferent nerves of the kidney are sparse.[5]

Glomerulus (Kidney Corpuscle)

The glomerulus includes a tuft of specialized capillaries attached to the mesangium, both of which are enclosed in a pouch-like extension of the tubule, which represents Bowman's capsule (Figs. 1.4 and 1.5). The capillaries, together with the mesangium, are covered by epithelial cells (podocytes), forming the visceral epithelium of Bowman's capsule. At the vascular pole, this is reflected to become the parietal epithelium of Bowman's capsule. At the interface between the glomerular capillaries and the mesangium on one side and the podocyte layer on the other side, the glomerular basement membrane (GBM) is developed. The space between both layers of Bowman's capsule represents the urinary space, which, at the urinary pole, continues as the tubule lumen.

On entering the tuft, the afferent arteriole immediately divides into several primary capillary branches, each of which gives rise to an anastomosing capillary network representing a glomerular lobule. In contrast, the efferent arteriole is already established inside the tuft by

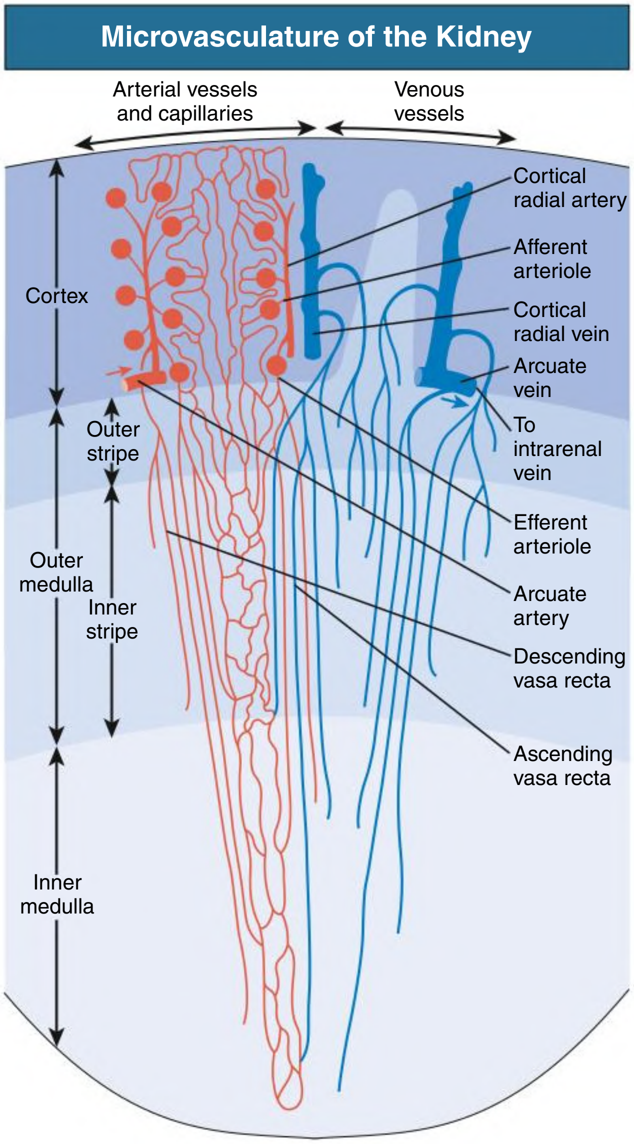

Fig. 1.3 Microvasculature of the Kidney. Afferent arterioles supply the glomeruli and efferent arterioles leave the glomeruli and divide into the descending vasa recta, which, together with the ascending vasa recta, form the vascular bundles of the renal medulla. The vasa recta ascending from the inner medulla all traverse the inner stripe within the vascular bundles, whereas most of the vasa recta from the inner stripe of the outer medulla ascend outside the bundles. Both types traverse the outer stripe as wide, tortuous channels.

confluence of capillaries from each lobule.[6] Thus the efferent arteriole has a significant intraglomerular segment located within the glomerular stalk.

Glomerular capillaries are a unique type of blood vessel made up of nothing but an endothelial tube (Figs. 1.6 and 1.7). A small stripe of the outer aspect of this tube directly abuts the mesangium; the major part bulges toward the urinary space and is covered by the GBM and the podocyte layer. This peripheral portion of the capillary wall represents the filtration area. The glomerular mesangium establishes the axis of a glomerular lobule to which the GBM, together with the glomerular capillaries, are attached.

Glomerular Basement Membrane

The GBM serves as the skeleton of the glomerular tuft. This membrane is a complexly folded sack with an opening at the glomerular hilum

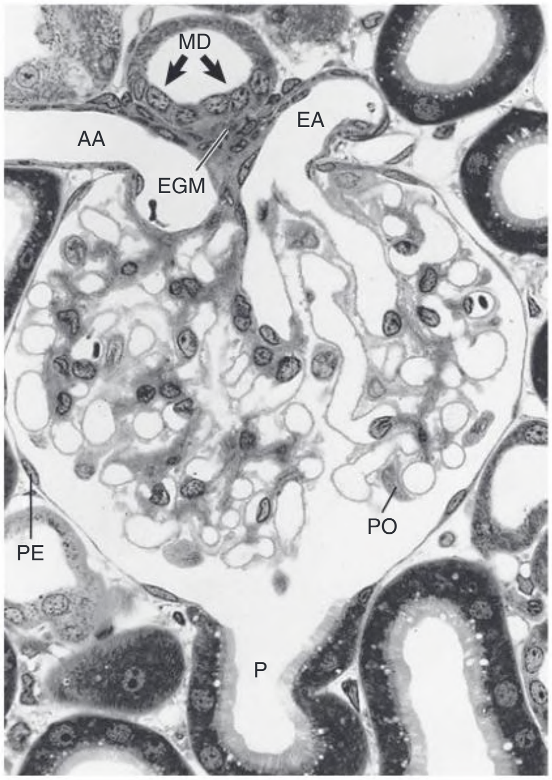

Fig. 1.4 Longitudinal Section Through a Glomerulus (Rat). At the vascular pole, the afferent arteriole *(AA)*, the efferent arteriole *(EA)*, the extraglomerular mesangium *(EGM)*, and the macula densa *(MD)* are seen. At the urinary pole, the parietal epithelium *(PE)* transforms into the proximal tubule *(P)*. *PO*, podocyte. (Light microscopy; magnification ×390.)

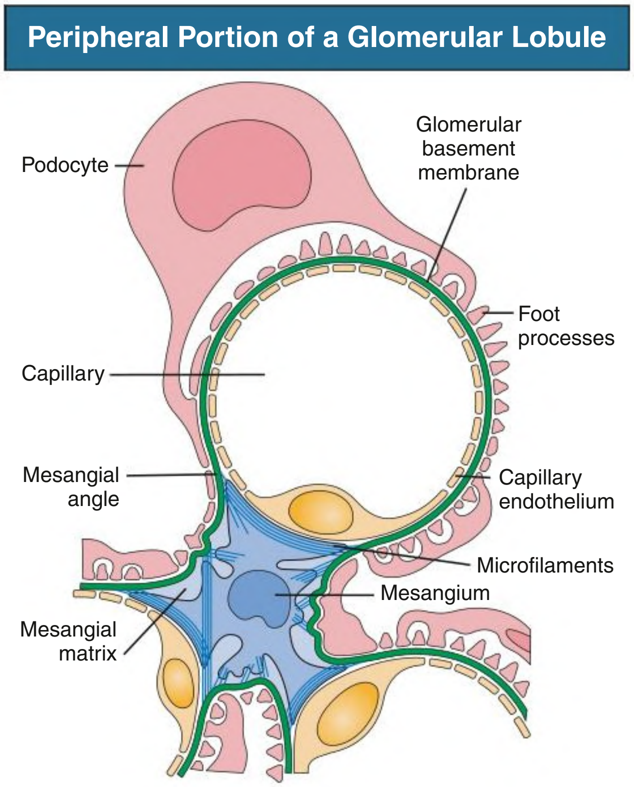

Fig. 1.5 Peripheral Portion of a Glomerular Lobule. This part shows a capillary, the axial position of the mesangium, and the visceral epithelium (podocytes). At the capillary-mesangial interface, the capillary endothelium directly abuts the mesangium.

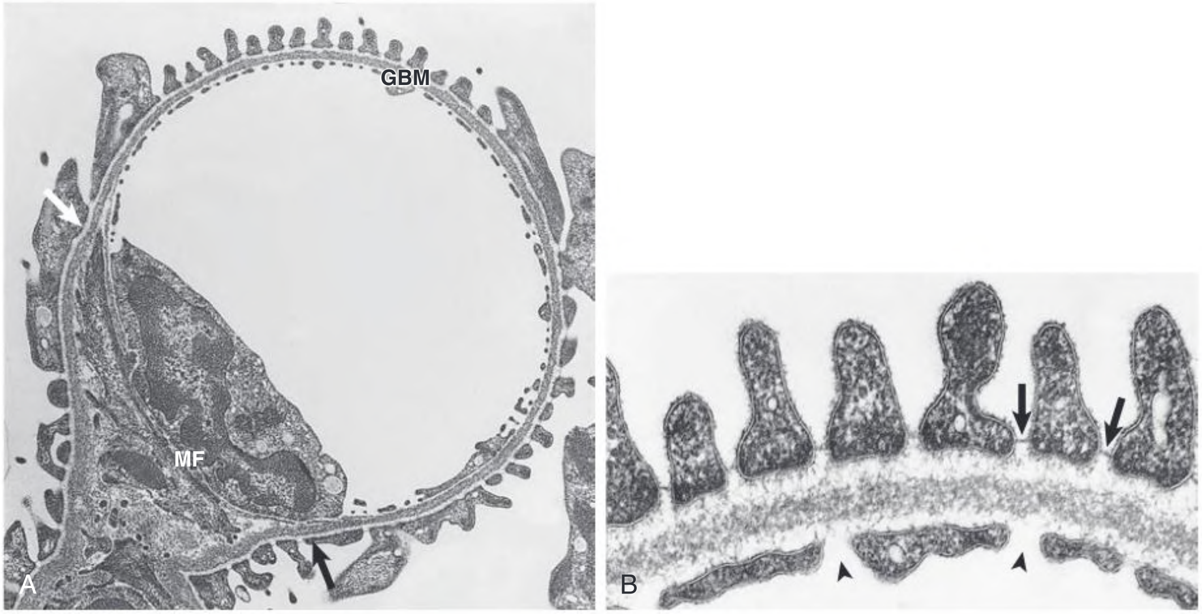

Fig. 1.6 Glomerular Capillary. (A) The layer of interdigitating podocyte processes and the glomerular basement membrane *(GBM)* do not completely encircle the capillary. At the mesangial angles *(arrows),* both deviate from a pericapillary course and cover the mesangium. Mesangial cell processes containing dense bundles of microfilaments *(MF)* interconnect the GBM and bridge the distance between the two mesangial angles. (B) Filtration barrier. The peripheral part of the glomerular capillary wall includes the endothelium with open pores *(arrowheads),* the GBM, and the interdigitating foot processes. The GBM shows a lamina densa bounded by the lamina rara interna and externa. The foot processes are separated by filtration slits bridged by thin diaphragms *(arrows).* (TEM; A, ×8770; B, ×50,440.)

(Fig. 1.8). The outer aspect of this GBM sack is completely covered with podocytes. The interior of the sack is filled with the capillaries and the mesangium. As a result, on its inner aspect, the GBM is in contact with either capillaries or the mesangium. At any transition between these two locations, the GBM changes from a convex pericapillary to a concave perimesangial course; the turning points are called *mesangial angles.* In electron micrographs, the GBM appears as a trilaminar structure, with a lamina densa bounded by two less dense layers, the lamina rara interna and lamina rara externa (see Fig. 1.6). Studies with freeze techniques reveal only one thick, dense layer directly attached to the bases of the epithelium and endothelium.[7]

The major components of the GBM include type IV collagen, laminin, and heparan sulfate proteoglycans similar to other basement membranes. Types V and VI collagen and nidogen (entactin) have also been demonstrated. Unique features of the GBM encompass a distinct spectrum of type IV collagen and laminin isoforms. The mature GBM consists of type IV collagen (with the major part made of α3, α4, and α5 chains, produced by podocytes, and the minor part made of α1, and α2 chains, produced by endothelial cells) and laminin 11, made of α5, β2, and γ1 chains.[8] Type IV collagen is the antigenic target in Goodpasture disease (see Chapter 25), and mutations in the genes of the α3, α4, and α5 chains of type IV collagen are responsible for Alport syndrome (see Chapter 48).

Current models depict the basic structure of the GBM as a three-dimensional network of type IV collagen.[7] The type IV collagen monomer consists of a triple helix that is 400 nm in length, with a large, noncollagenous globular domain at its C-terminal end called NC1. At the N terminus, the helix possesses a triple helical rod 60 nm long: the 7S domain. Interactions between the 7S domains of two triple helices or the NC1 domains of four triple helices allow type IV collagen monomers to form dimers and tetramers. In addition, triple helical strands interconnect by lateral associations through binding of NC1 domains to sites along the collagenous region. This network is complemented by an interconnected network of laminin 11, resulting in a flexible, nonfibrillar polygonal assembly that provides mechanical strength and elasticity to the basement membrane and serves as a scaffold for alignment of other matrix components.[9,10]

The electronegative charge of the GBM mainly results from the presence of polyanionic proteoglycans. The major proteoglycans of the GBM are heparan sulfate proteoglycans, including perlecan and agrin. Proteoglycan molecules aggregate to form a meshwork that is kept well hydrated by water molecules trapped in the interstices of the matrix.

Mesangium

Three major glomerular cell types are all in close contact with the GBM: mesangial cells (MCs), endothelial cells, and podocytes. The mesangial/endothelial/podocyte cell ratio is 2:3:1 in the rat. The mesangial cells and mesangial matrix establish the glomerular mesangium.

Mesangial cells. MCs are irregular in shape, with many processes extending from the cell body toward the GBM (see Figs. 1.5 and 1.6). In these processes, dense assemblies of microfilaments are found, containing α-smooth muscle actin, myosin, and α-actinin.[11] The processes are attached to the GBM directly or through the interposition of microfibrils. The GBM represents the effector structure of mesangial contractility. MC-GBM connections are found throughout the mesangium-GBM interface, but they are especially prominent at the turning points of the GBM infoldings (mesangial angles). A new adhesion complex has been uncovered at these sites,[12] consisting of nephronectin deposited by podocytes into the GBM and α8 β1-integrins enriched in the tips of MC processes inserting to the GBM.

The folding pattern of the GBM is permanently challenged by the expansile forces of the high intraglomerular perfusion pressure. Centripetal MC contraction balances the expansile forces. Thus the

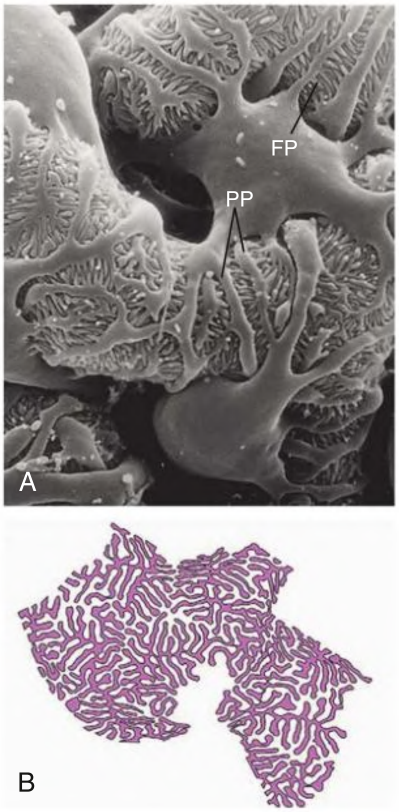

Fig. 1.7 **Branching Pattern of Podocyte Foot Processes (Rat).** (A) Scanning electron micrograph (SEM) showing the urinary side of the podocyte cover of a glomerular capillary consisting of cell bodies, large primary processes *(PP),* and interdigitating foot processes *(FP)* separated by the filtration slits. (B) Drawing of the basal aspect of the FP-branching pattern as seen by block-face SEM. A fully homogenous branching pattern of foot processes attaches to the glomerular basement membrane (GBM), which may be compared with a pattern of interdigitating filopodia connected by adherens junctions. The high degree of branching (not seen from the luminal aspect) provides a high degree of adaptability to area changes of the underlying GBM. (From Ichimura K, Kakuta S, Kawasaki Y, et al. Morphological process of podocyte development revealed by block-face scanning electron microscopy. *J Cell Sci.* 2017;130:132–142.)

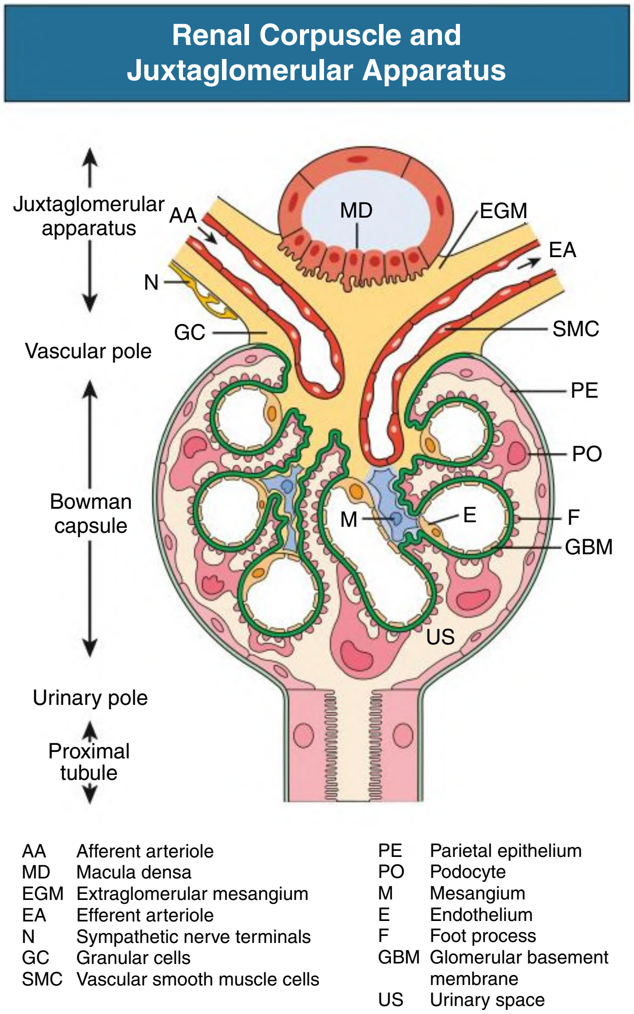

Fig. 1.8 **Renal Corpuscle and Juxtaglomerular Apparatus.** (Modified from Kriz W, Kaissling B. Structural organization of the mammalian kidney. In Seldin DW, Giebisch G, eds. *The Kidney,* 3rd ed. Lippincott Williams & Wilkins: 2000;587–654.)

folding pattern of the GBM, including the complex convolutions of glomerular capillaries, are maintained by mesangial cells.

MCs possess a great variety of receptors, including those for angiotensin II (Ang II), vasopressin, atrial natriuretic factor, prostaglandins, and other growth factors (tumor growth factor [TGF]-β, platelet-derived growth factor [PDGF], epidermal growth factor [EGF], and connective tissue growth factor [CTGF]).[13]

Mesangial matrix. The mesangial matrix fills the highly irregular spaces between the mesangial cells and the perimesangial GBM, anchoring the mesangial cells to the GBM.[6] Many common extracellular matrix proteins have been demonstrated within the mesangial matrix, including collagen types IV, V, and VI and microfibrillar protein components, such as fibrillin and microfibril-associated glycoprotein. The matrix also contains several glycoproteins, most abundantly fibronectin, and several types of proteoglycans.

Endothelium

Glomerular endothelial cells consist of cell bodies and peripherally located, attenuated, and highly porous cytoplasmic sheets (see Figs. 1.5 and 1.6). Glomerular endothelial pores lack diaphragms, which are encountered only in the endothelium of the final tributaries to the efferent arteriole.[6] The round to oval pores have a diameter of 50 to 100 nm. A layer of membrane-bound and loosely attached molecules (called the glycocalyx) covers the entire luminal surface including, as sieve plugs, the endothelial pores.[14] Because of a great variety of polyanionic glycoproteins (including podocalyxin), the endothelium has a strong negative charge, establishing an exit barrier for negatively charged macromolecules like albumin. Endothelial cells actively participate in processes controlling coagulation and inflammation and have receptors for vascular endothelial growth factor (VEGF), angiopoietins, and TGF-β1. They synthesize and release PDGF-B, endothelin-1, and nitric oxide, among others.[15]

Visceral Epithelium (Podocytes)

The visceral epithelium of Bowman's capsule includes highly differentiated cells called the podocytes (see Fig. 1.7; see also Fig. 1.5). In the developing glomerulus, podocytes have a simple polygonal shape. In humans, podocytes complete mitosis during prenatal life. Differentiated podocytes are unable to replicate; consequently, lost podocytes cannot be replaced in the adult. All efforts of the last decade to find progenitor cells that might migrate into the tuft and replace lost podocytes have failed.

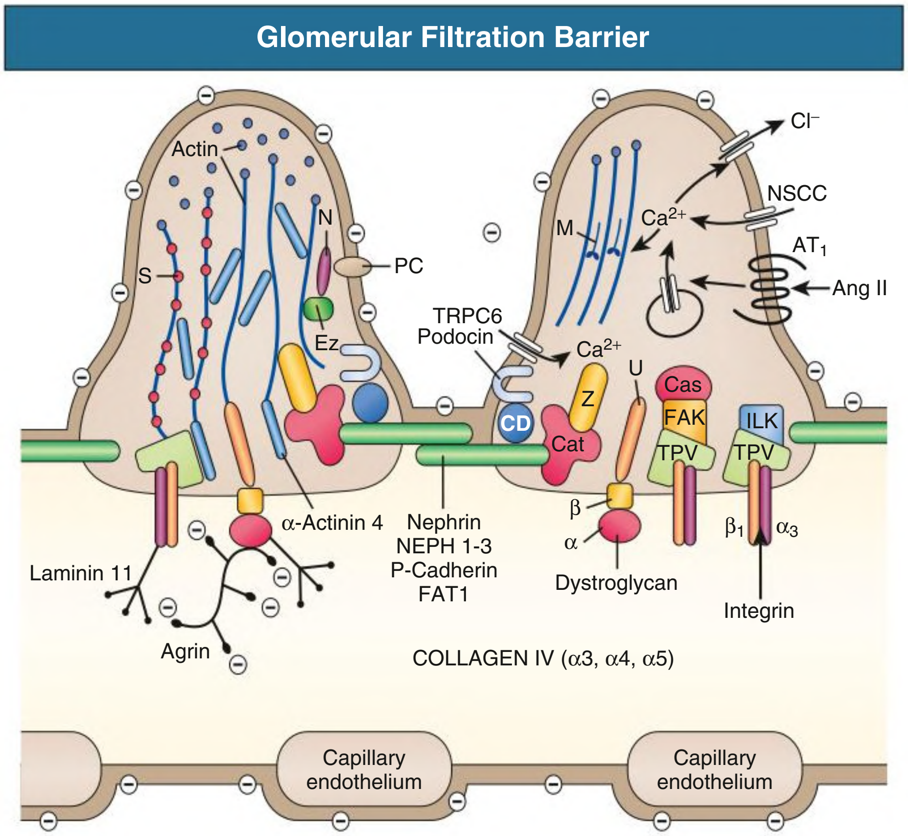

Fig. 1.9 Glomerular Filtration Barrier. Two podocyte foot processes bridged by the slit membrane, the glomerular basement membrane (GBM), and the porous capillary endothelium are shown. The surfaces of podocytes and of the endothelium are covered by a negatively charged glycocalyx containing the sialoprotein podocalyxin (PC). The GBM is mainly composed of type IV collagen (α3, α4, and α5), laminin 11 (α5, β2, and γ1 chains), and the heparan sulfate proteoglycan agrin. The slit membrane represents a porous proteinaceous membrane composed of (as far as is known) nephrin, NEPH 1-3, P-cadherin, and FAT1. The actin-based cytoskeleton of the foot processes connects to both the GBM and the slit membrane. Regarding the connections to the GBM, β1α3 integrin dimers specifically interconnect the TPV complex (talin, paxillin, vinculin) to laminin 11; the β- and α-dystroglycans interconnect utrophin to agrin. The slit membrane proteins are joined to the cytoskeleton by various adapter proteins, including podocin, zonula occludens protein 1 (ZO-1; Z), CD2-associated protein (CD), and catenins (Cat). Among the nonselective cation channels (NSCC), TRPC6 associates with podocin (and nephrin, not shown) at the slit membrane. Only the angiotensin II (Ang II) type 1 receptor (AT_1) is shown as an example of the many surface receptors. *Cas,* p130Cas; *Ez,* ezrin; *FAK,* focal adhesion kinase; *ILK,* integrin-linked kinase; *M,* myosin; *N,* Na^+-H^+ exchanger regulatory factor (NHERF2); *S,* synaptopodin. (Modified from Endlich KH, Kriz W, Witzgall R. Update in podocyte biology. *Curr Opin Nephrol Hypertens.* 2001;10:331–340.)

Podocytes have a voluminous cell body that floats within the urinary space, separated from the GBM by a subpodocyte space.[16] The cell bodies give rise to primary processes that fall apart into foot processes (FPs) that fix the cells to the capillaries (i.e., to the GBM). Sporadic FPs may also arise directly from the cell body. The FPs of neighboring podocytes regularly interdigitate with each other, leaving meandering slits (filtration slits) between them that are bridged by a complex extracellular structure known as the *slit diaphragm* (SD) that may be seen as a modified adherens junction (Fig. 1.9; see also Figs. 1.6 and 1.7). Traditional scanning electron micrograph (SEM) pictures (see Fig. 1.7A) do not convey the correct pattern of how FPs interdigitate and adhere to the GBM. As seen by block-face SEM (see Fig. 1.7B), individual FP may terminate with a final branching, and primary processes fall off into basal ridges that actually are also FPS.[17] Thus the interdigitating FP pattern as it adheres to the GBM is completely homogenous, forming a uniform cover of interdigitating filopodia. This is of utmost importance to understand the adaptive mobility of this cover (see later).

The cell body contains a prominent endoplasmic reticulum and Golgi system and a well-developed endocytotic and autophagic machinery. A complex cytoskeleton accounts for the shape of the cells. In the cell body and the primary processes, microtubules and intermediate filaments (vimentin, desmin) dominate. Within the FPs, microfilaments (β-actin) form prominent U-shaped bundles arranged in the longitudinal axis of two successive FPs in an overlapping pattern. Above, the bends of these bundles are linked to the microtubules of the primary processes; peripherally, these bundles terminate in the dense cytoplasm associated with the sole plates, being part of the anchoring system of the FPs to the GBM (see later). In addition, the cell body and the FPs have a well-developed subplasmalemmal actin network that has intimate contact to the anchor line of the SD and diffusely to the actin bundles of FPs. Multiple actin-associated proteins, including α-actinin-4 and synaptopodin, establish the specific cytoskeleton in podocytes.[18] FPs do not contain myosin II[19]; thus the actin fibers in FPs are not part of a contractile system. Instead, the complex cytoskeleton of FPs supports the attachment of FPs to the GBM and the maintenance of their interdigitating pattern.[20]

The luminal membrane contains a great variety of receptors (see later) and, together with the luminal surface of the SD, it is covered by a thick surface coat that is rich in sialoglycoproteins, including podocalyxin and podoendin, accounting for the high negative surface charge of the podocytes.

The abluminal cell membrane consists of a narrow band of lateral cell membrane extending from the SD to the GBM and, most importantly, the soles of the FPs abutting to the GBM. A complex anchoring system connects the cytoskeleton of the FPs to the GBM. Two systems are known: (1) $\alpha3\beta1$ integrin dimers interconnect the cytoplasmic focal adhesion proteins vinculin, paxillin, and talin with the $\alpha3$, $\alpha4$, and $\alpha5$ chains of type IV collagen and laminin 521; and (2) β-α-dystroglycans interconnect the cytoplasmic adapter protein utrophin with agrin and laminin $\alpha5$ chains in the GBM.[9]

The junctional connection of podocyte FPs by the SD bridging the filtration slits is complex and unique. Filtration slits have a constant width of about 30 to 40 nm: thus SD has to connect the FPs over a considerable distance. By transmission electron microscopy (TEM), the SD shows up as a single dark line in cross sections and in an en-face view as a homogenous network of fibrillar structures interconnecting both membranes. Combined tannic acid and glutaraldehyde fixed tissue reveals, in en-face view, a zipper-like structure with a row of pores approximately 14×2 nm on either side of a central bar. The transmembrane proteins that establish the slit membrane and its connection to the actin cytoskeleton of the FPs include nephrin, P-cadherin, FAT1, NEPH 1-3, podocin, and CD2AP[21] (see Fig. 1.9).

Podocytes contain many surface receptors and ion channels (see Fig. 1.9).[21] Among the transient receptor potential (TRP) cation channels, TRPC5 and TRPC6 have received much attention.[22–24] The major target of this signaling orchestra is the cytoskeleton (see later). Other receptors, such as for TGF-β, FGF-2, and other cytokines/chemokines, are involved in synthesis functions (GBM components) or in the development of podocyte diseases.[21] Megalin is a multiligand endocytotic receptor and the major antigen of Heymann nephritis in the rat[25] but is not present in humans.[13] However, human podocytes express phospholipase A-2 receptor, which is the autoantigen targeted in the majority of patients with idiopathic membranous nephropathy.

Podocytes, by paracrine and autocrine signaling, regulate the interplay with endothelial cells and MCs; during development, they have the exclusive commandership in building a glomerulus. VEGF, angiopoietins, and PDGF, among others, are of crucial importance for the homeostatic maintenance of the tuft.[26]

Function and Maintenance of the Filtration Barrier

Filtration pressure and expansion. Traditionally, the high transmural hydrostatic pressure gradients necessary for filtration have been considered as the main challenge to the filtration barrier. Podocyte FPs were considered as pericyte processes that counteracted variations and derailments in perfusion pressures. This view has been challenged because of the discovery that podocytes are chiefly lost by detachment from the GBM as viable cells. It seems self-contradictory that FPs, which need their cytoskeleton to continually adapt their pattern of attachment to the GBM (see later), would simultaneously function as contractile pericyte-like processes, counteracting the expansion of the GBM by increasing their tone. This agrees with the finding that FPs do not contain any myosin.[19] Consequently, the principal burden for counteracting transmural pressure gradients (i.e., for developing wall tension) falls instead on the GBM.[27]

As previously described, the GBM is an elastic membrane that expands or shrinks in surface area with increasing or decreasing transmural hydrostatic pressure, respectively. Its expansion decreases with increasing pressure and is limited. Thus podocytes situated downstream of the GBM are protected against sudden pressure rises. A situation in which the pressure is high enough to disrupt the GBM has never been reported.

Expansion of the GBM affords the immediate coordinated increase in the cover by interdigitated FPs, thus the FPs and the SD have to increase correspondingly (and vice versa when pressure decreases). The ability to make such acute adaptions has previously been shown in the isolated perfused kidney. It is suggested that the changes in FP length occur by actin polymerization/depolymerization; the mechanism underlying changes in SD length is unknown.[20,27]

Filtrate flow and shear stress. The flow of the filtrate through the filtration barrier represents by far the highest extravascular fluid flow in the body. It consists of the outflow from glomerular capillaries, through the GBM, and into Bowman's space. This latter step creates a problem: in contrast to the exit of filtrate from capillaries, where flow presses the endothelium against the basement membrane, its entry into Bowman's space tends to separate the podocytes from the GBM. This danger explains the high density of integrin-connections of FPs to the GBM.

The shear stress depends on the flow rate and the geometry of the channel; the narrower the channel or the higher the flow velocity, the higher the shear stress. In rats, the filtrate flow amounts to 30 nL/min, creating a shear stress to the FPs within the filtration slit as high as 8 Pa.[28] Much lower values of shear stress to the podocyte cell bodies (0.5 Pas when they come to lie within the urinary orifice) lead to the detachment of podocytes from the GBM and loss with the urine flow as viable cells; this has repeatedly been shown by light microscopy (LM) and TEM in experimental disease models with podocyte loss.[20] Moreover, a high sensitivity of podocytes to shear stress has been shown in cell culture studies.

This led to a new view of the relevance of the slit membrane (in addition to its barrier function; see later). Shear stress tends to lead to deformations of the lateral walls of FPs, thus widening the slit. The interconnection of both opposite FPs by the SD at the narrowest site of the slit is ideally positioned to counteract these destabilizing forces. The SD uses the shear stress against one side of the slit to balance the shear stress against the opposite site. This means that during filtrate flow the SD is permanently under tension, which counteracts the shear stress to both sides of the slit.[20] This hypothesis is supported by the observation that the loss of the SD connection between adjacent FPs represents the earliest failure that starts the detachment of a podocyte (see Pathology).

Barrier function. Filtrate flow through the barrier occurs along an extracellular route, including the endothelial pores, GBM, and slit diaphragm (see Figs. 1.6 and 1.9). All these components are quite permeable for water, resulting in a high permeability for water, small solutes, and ions. On the other hand, the barrier is fairly tight for macromolecules and selective for size, shape, and charge.[21] The charge selectivity of the barrier results from the dense accumulation of negatively charged molecules on the surface coat (glycocalyx) of endothelial cells. Because most plasma proteins, including albumin, are negatively charged, their entrance into the barrier is slowed but not eliminated. Albumin that penetrates the barrier is filtered and reabsorbed in the proximal tubule.

The size/shape selectivity seems to be established by the SD.[14] Uncharged macromolecules up to an effective radius of 1.8 nm pass freely through the filter. Larger components are increasingly restricted and are totally excluded at effective radii of greater than 4 nm. The fate of components that are captured inside the SD is unknown; transcytosis through podocytes has been suggested.[29] Plasma albumin has an effective radius of 3.6 nm; without the repulsion from the negative charge, plasma albumin would pass through the filter in considerable amounts.

Moeller[30] has proposed an electrophoretic mechanism for the repulsion and exclusion of plasma proteins from the glomerular filter based on the flow of the filtrate through the charged filter, creating a streaming potential.[31] This electrical field is negatively charged on the urinary side of the glomerular filter compared with the capillary side by approximately −0.05 mV/10 mm Hg filtration pressure. Thus albumin molecules (and most plasma proteins, because virtually all are negatively charged) that approach the filter will be exposed to an electrophoretic force driving them back toward the capillary lumen. The charm of this hypothesis consists of its independence from any structural impediment to filtration. The barrier actually consists of a filtration-dependent potential difference; without sufficient convective flow of filtrate, the barrier will become permeable.[31]

Pathology. The hypothesis that the mechanical interconnection of the FPs by the SD is critical for maintenance of an intact barrier and is vulnerable to the physical challenges of filtration is supported by the pathologic changes. The loss of the SD connection between adjacent FPs initiates the detachment of podocytes[20] and the loss of local control of filtrate flow.

Unchanneled filtrate flow through such leaks will exert unbalanced shear stress on the FPs starting local detachment of FPs. Repair of such leaks is difficult in the face of ongoing filtrate flow, accounting for the observation that the damage tends to progress.

Taken together, the layer of interdigitating FPs interconnected by the SD regulates the entry of the filtrate flow into Bowman's space by channeling the flow through the filtration slits. The geometry of the slits is maintained against the shear forces to both opposite FPs through the interconnection of opposing FPS by the SD. Loss of the junctional connection is detrimental because it opens leaks for uncontrolled filtrate flow with the tendency to increase the leaks.[32] This situation also accounts for the fact that podocytes cannot undergo cytokinesis and, when lost, cannot be replaced by progenitors immigrating from outside.

Parietal Epithelium

The parietal epithelium of Bowman's capsule consists of squamous epithelial cells resting on a basement membrane (see Figs. 1.4 and 1.8). The flat cells are filled with bundles of actin filaments running in all directions. In contrast to the GBM, the parietal basement membrane has several proteoglycan-dense layers that, in addition to type IV, contain type XIV collagen. The predominant proteoglycan of the parietal basement membrane is a chondroitin sulfate proteoglycan.[33] A niche of epithelial stem cells may reside within the parietal epithelium at the transition to the proximal tubule and/or to the visceral epithelium,[34,35] but unequivocal evidence is lacking.

Renal Tubule

The renal tubule consists of distinct segments: a proximal tubule (convoluted and straight portions), an intermediate tubule, a distal tubule (straight and convoluted portion), a connecting tubule (CNT), and the collecting duct (see Figs. 1.1 and 1.2; Table 1.1).[1,2,33] Henle's loop includes the straight part of the proximal tubule (representing the thick descending limb), the thin descending and the thin ascending limbs (both thin limbs together represent the intermediate tubule), and the thick ascending limb (representing the straight portion of the distal tubule), which includes the macula densa. The CNT connects the nephron to the collecting duct system.

Tubular epithelium consists of a single layer of cells anchored to a basement membrane. The cells have multiple transport functions and show numerous structural adaptations to their special roles. They are connected apically by a junctional complex consisting of a belt-like tight junction (zonula occludens), a belt-like adherens junction (zonula adherens), and, at some sites, a desmosome. The tight junction represents a structural border between the luminal and the basolateral cell membrane domain, preventing movements of membrane-associated transport proteins between the two domains. Two different pathways through the epithelium exist (Fig. 1.10): a transcellular pathway, including the transport across the luminal and the basolateral cell membrane and through the cytoplasm, and a paracellular pathway through the junctional complex and the lateral intercellular spaces. The functional characteristics of the paracellular transport are determined by the tight junction, which differs markedly in its elaboration in the various tubular segments. The tight junctional proteins responsible for the specific permeability properties are occludin and especially the claudins.[36] Transcellular transport is determined by apical and basolateral channels, carriers, and transporters. The various nephron segments differ markedly in function, distribution of transport proteins, and responsiveness to hormones and drugs such as diuretics. The cell surface area of the plasmalemmal compartments carrying the transport systems is extensively enlarged in many tubule cells: namely, by microvilli at the luminal membrane domain, by lamellar folds of the basolateral membrane interdigitating with those of the neighboring cells (interdigitations), or by lamellar folds of the basal cell membrane invaginating into the own cell (invaginations).

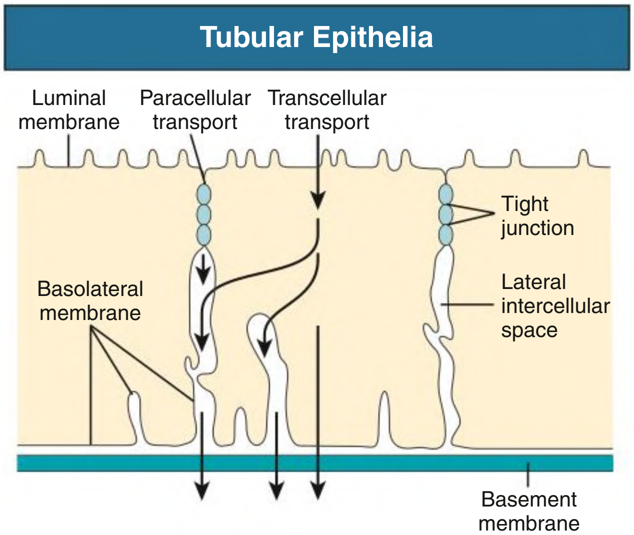

Fig. 1.10 Tubular Epithelia. Transport across the epithelium may follow two routes: transcellular, across luminal and basolateral membranes, and paracellular, through the tight junction and intercellular spaces.

Proximal Tubule

The proximal tubule reabsorbs the bulk of filtered water and solutes (Fig. 1.11) and is generally subdivided into three segments (known as S_1, S_2, S_3) that differ considerably in cellular organization and, consequently, also in function.[37] The proximal tubule has a prominent brush border and extensive interdigitation by basolateral cell processes. These lateral cell interdigitations extend up to the leaky tight junction, thus considerably increasing the tight junctional belt in length and providing a greatly increased passage for the passive transport of ions. Proximal tubule cells have large prominent mitochondria intimately associated with the basolateral cell membrane where the Na^+,K^+–adenosine triphosphatase (ATPase) is located; this machinery is the molecular mechanism initiating numerous secondary transcellular transport processes. The luminal transporter for Na^+

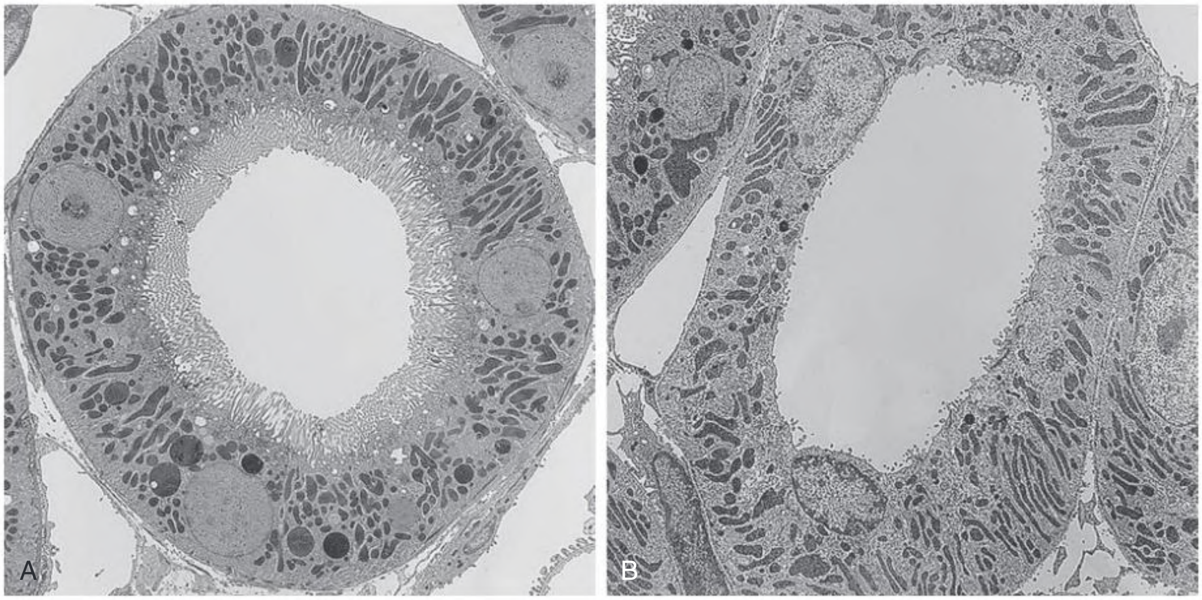

Fig. 1.11 **Tubules of the Renal Cortex.** (A) Proximal convoluted tubule is equipped with a brush border and a prominent vacuolar apparatus in the apical cytoplasm. The rest of the cytoplasm is occupied by a basal labyrinth consisting of large mitochondria associated with basolateral cell membranes. (B) Distal convoluted tubule also has interdigitated basolateral cell membranes intimately associated with large mitochondria. In contrast to the proximal tubule, however, the apical surface is amplified only by some stubby microvilli. (TEM; A, ×1530; B, ×1830.)

reabsorption specific for the proximal tubule is the Na^+-H^+ exchanger (NHE3) that is located in the plasma membrane of the apical microvilli and accounts for reabsorption of most of the filtered sodium. Other sodium coupled transporters in the microvillous membrane are the sodium-glucose cotransporters SGLT2 and SGLT1 and several sodium-phosphate cotransporters. The abundance of channel protein aquaporin 1 in the apical microvillous membrane and the basolateral cell membrane accounts for the high hydraulic permeability for water of this epithelium. An apical tubulovesicular compartment is part of the prominent endosomal-lysosomal system and is responsible for the reabsorption of macromolecules (active peptides, polypeptides, and proteins such as albumin) that have passed the glomerular filter.

The S_3 and portions of the S_2 segments are engaged in many secretory processes of toxic substances and drugs via organic anion transporters and an organic cation transporter (OCT). Proximal tubule cells are electrically coupled by gap junctions (Nexus).

Intermediate Tubule

The intermediate tubule includes the thin portion of Henle's loop, thin descending and (only in long loops) thin ascending limbs (Fig. 1.12; see also Fig. 1.2). Thin descending limbs of short loops and those of long loops are equipped with different epithelia. The thin descending limbs of short loops have an extremely flat epithelium that, in its beginning part, is permeable to water, and, in its distal part, contains the urea transporter UT-A2 responsible for the uptake of urea coming up from the inner medulla (urea recycling).[38] The thin descending limbs of long loops, in their initial parts within the inner stripe, are quite permeable to Na^+ and Cl^- (uptake from the interstitial space) and have a considerable Na^+,K^+-ATPase activity, the relevance of which is unknown. Along their course down through the inner medulla they lose their salt permeability becoming permeable to water. The transition to the ascending thin limb epithelium occurs already before the bend. The ascending thin limbs have a heavily interdigitated epithelium highly permeable for ions releasing them into the interstitial space but impermeable for water. The relevance of transport functions of the thin limbs for the generation of the osmotic medullary gradient is debated.

Distal Straight Tubule (Thick Ascending Limb of Henle's Loop)

The thick ascending limb of Henle's loop is often called the diluting segment. It is water impermeable but reabsorbs considerable amounts of sodium and chloride, resulting in the separation of salt from water. The salt is trapped in the medulla (see Fig. 1.12), whereas the water is carried away into the cortex, where it may return into the systemic circulation. The specific transporter for Na^+ reabsorption in this segment is the Na^+-K^+-$2Cl^-$ cotransporter 2 (NKCC2),[39] which is specifically inhibited by loop diuretics such as furosemide. This transporter is inserted in the luminal membrane, which is amplified by only solitary microvilli. The tight junctions of the thick ascending limb are elongated by lateral interdigitation of the cells. They have a comparatively low overall permeability; however, they contain the protein Claudin 16 for paracellular reabsorption of divalent ions, notably of magnesium. The cells are heavily interdigitated by basolateral cell processes, which are associated with large mitochondria supplying the energy for the transepithelial transport. The cells synthesize the Tamm-Horsfall protein and release it into the tubular lumen. This protein may help prevent the formation of kidney stones. A short distance before the transition to the distal convoluted tubule, the thick ascending limb contains the macula densa, which adheres to the glomerulus of the same nephron (see Juxtaglomerular Apparatus).

Distal Convoluted Tubule

The epithelium exhibits the most extensive basolateral interdigitation of the cells and the greatest numerical density of mitochondria compared with all other nephron portions (see Fig. 1.11). Apically, the cells are equipped with numerous solitary microvilli. The specific Na^+ transporter of the distal convoluted tubule is the luminal Na^+-Cl^- cotransport system (NCC), which can be inhibited by the thiazide diuretics. Magnesium is reabsorbed via the transient receptor potential channel melastatin subtype 6 (TRPM6) in the luminal membrane[40]

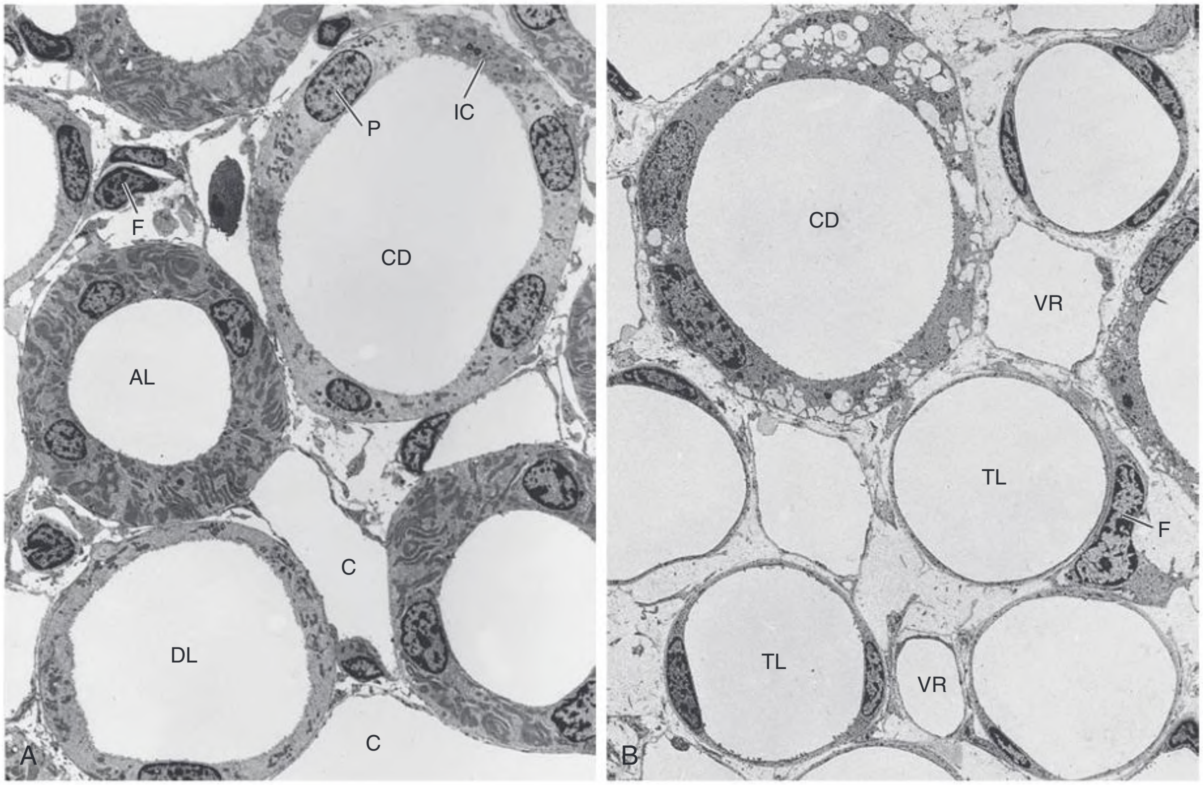

Fig. 1.12 Tubules in the Medulla. (A) Cross-section through the inner stripe of the outer medulla shows a descending thin limb of a long Henle loop *(DL),* the medullary thick ascending limbs of Henle *(AL),* and a collecting duct *(CD)* with principal cells *(P)* and intercalated cells *(IC).* (B) In the inner medulla cross-section, thin descending and ascending limbs *(TL),* a collecting duct *(CD),* and vasa recta *(VR)* are seen. *C,* Peritubular capillaries; *F,* fibroblast. (TEM; A, ×990; B, ×1120.)

and, along the paracellular route, through the tight junctional proteins Claudin 16 and 19.

COLLECTING DUCT SYSTEM

The collecting duct system (see Fig. 1.2) includes the CNT, cortical ducts, and medullary collecting ducts (CDs). The embryologic origin of the CNT, which is interposed between the distal convoluted tubule and the CD, is unclear whether it derives from the nephron anlage or the ureteral bud. Two nephrons may join at the level of the CNT, forming an arcade. Two types of cell establish the CNT: the CNT cell, which is specific to the CNT, and the intercalated (IC) cell, which is also present in varying amounts in the distal convoluted tubule and in the collecting duct. The CNT cells are similar to the CD cells in cellular organization. Both cell types share sensitivity to vasopressin (antidiuretic hormone [ADH]; see later). The amiloride-sensitive epithelial sodium channel, ENaC, and the epithelial calcium channel, TRPV5, are located in the apical membrane, being already present in the distal convoluted tubule and extending into the CNT.

Collecting Ducts

The CDs (see Fig. 1.12) may be subdivided into cortical and medullary ducts, and the medullary ducts into an outer and inner portion; the transitions are gradual. Like the CNT, the CDs are lined by two types of cell: CD cells (principal cells) and IC cells. The IC cells decrease in number as the CD descends into the medulla and are absent from the inner medullary CDs.

The CD cells (Fig. 1.13A) increase in size toward the tip of the papilla. The basal cell membrane amplifies by lamellar invaginations into the cell (basal infoldings). The tight junctions have a large apical-basal depth, and the apical cell surface has a prominent glycocalyx. These cells contain an apical shuttle system for aquaporin 2 under the control of vasopressin, allowing the water permeability of the CDs to switch from very low to permeable.[41] A luminal amiloride-sensitive Na^+ channel is involved in the responsiveness of cortical CDs to aldosterone. The terminal portions of the CD in the inner medulla express the urea transport system UT-A1, which, in an antidiuretic hormone (ADH)-dependent fashion, accounts for the recycling of urea, a process that is crucial in the urine-concentrating mechanism.[42,43]

The second cell type, the IC cell (see Fig. 1.13B), is present in both the CNT and the CD. There are at least two types of IC cells, designated A and B cells, distinguished on the basis of structural, immunocytochemical, and functional characteristics. Type A cells express a H^+-ATPase at their luminal membrane; they secrete protons. Type B cells express the H^+-ATPase at their basolateral membrane; they secrete bicarbonate ions and reabsorb protons.[44]

The CDs are the final regulators of fluid and electrolyte balance, playing important roles in the handling of Na^+, Cl^-, and K^+, as well as in acid-base homeostasis. The responsiveness of the CDs to vasopressin enables an organism to live in arid conditions, allowing it to produce a concentrated urine and, if necessary, a dilute urine.

JUXTAGLOMERULAR APPARATUS

The juxtaglomerular apparatus (JGA) includes the macula densa, the extraglomerular mesangium, the terminal portion of the afferent arteriole with its renin-producing granular cells (also often called

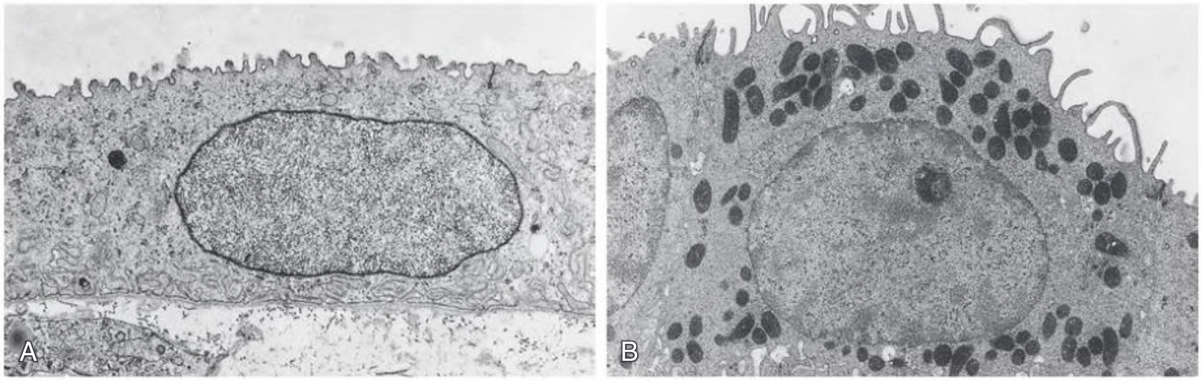

Fig. 1.13 Collecting Duct Cells. (A) Principal cell (collecting duct [CD] cell) of a medullary collecting duct. The apical cell membrane bears some stubby microvilli covered by a prominent glycocalyx; the basal cell membrane forms invaginations. Note the deep tight junction. (B) Intercalated cells, type A. Note the dark cytoplasm (dark cells) with many mitochondria and apical microfolds; the basal membrane forms invaginations. (TEM; A, ×8720; B, ×6970.)

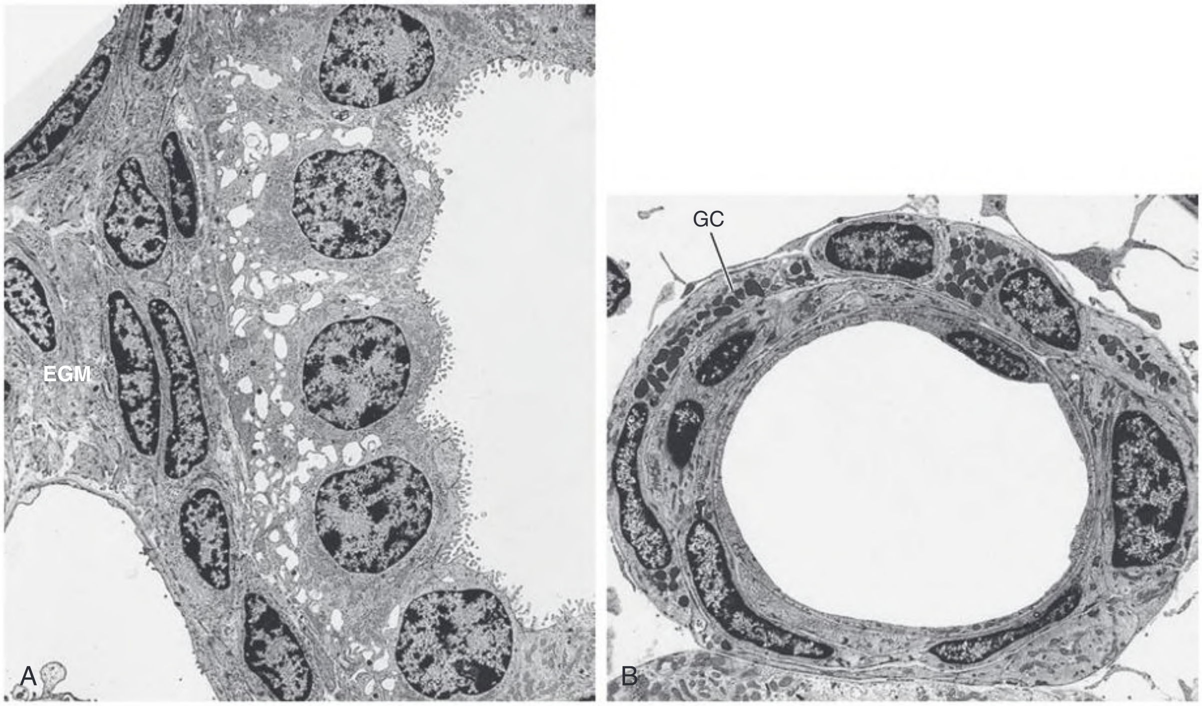

Fig. 1.14 Juxtaglomerular Apparatus. (A) Macula densa of a thick ascending limb of Henle. The cells have prominent nuclei and lateral intercellular spaces. Basally, they attach to the extraglomerular mesangium *(EGM).* (B) Afferent arteriole near the vascular pole. Several smooth muscle cells are replaced by granular cells *(GC)* containing accumulations of renin granules. (TEM; A, ×1730; B, ×1310.)

juxtaglomerular cells), and the beginning portions of the efferent arteriole (see Fig. 1.8).

The macula densa is a plaque of specialized cells in the wall of the thick ascending limb of Henle at the site where the limb attaches to the extraglomerular mesangium of the parent glomerulus (Fig. 1.14A; see also Fig. 1.4). The most obvious structural feature is the narrowly packed cells with large nuclei, which account for the name macula *densa.* The cells are anchored to a basement membrane, which blends with the matrix of the extraglomerular mesangium. The cells are joined by tight junctions with very low permeability and have prominent lateral intercellular spaces. The width of these spaces varies under different functional conditions.[1] The most conspicuous immunocytochemical difference between macula densa cells and other epithelial cells of the nephron is the high content of neuronal nitric oxide synthase and cyclooxygenase-2 in macula densa cells.[45,46]

The basal aspect of the macula densa is firmly attached to the extraglomerular mesangium, a solid complex of cells and matrix penetrated by neither blood vessels nor lymphatic capillaries. As with the mesangial cells proper, extraglomerular MCs are heavily branched. Their processes are interconnected by gap junctions, contain prominent bundles of microfilaments, and are connected to the basement membrane of Bowman's capsule and to the walls of both glomerular arterioles. As a whole, the extraglomerular mesangium interconnects all structures of the glomerular entrance.[6]

The granular cells are assembled in clusters within the terminal portion of the afferent glomerular arteriole (see Fig. 1.14B), replacing ordinary smooth muscle cells. "Granular" refers to the specific cytoplasmic granules in which renin, the major secretion product of these cells, is stored. Granular cells are the main site of the body where renin is secreted. Renin release occurs by exocytosis into the surrounding interstitium. Granular cells are connected to extraglomerular MCs, adjacent smooth muscle cells, and endothelial cells by gap junctions and are densely innervated by sympathetic nerve terminals. Granular cells are modified smooth muscle cells; under conditions requiring enhanced renin synthesis (e.g., volume depletion, renal artery stenosis), additional smooth muscle cells located upstream in the wall of the afferent arteriole may transform into granular cells.

The JGA serves two different functions: it regulates the flow resistance of afferent arterioles in the so-called tubule-glomerular feedback mechanism, and it participates in the control of renin synthesis and release from granular cells in the afferent arteriole. Renin release from granular cells is the major source of systemic angiotensin II and thus plays an essential role in controlling extracellular volume and blood pressure, whereas the vasoconstriction of the afferent arteriole locally serves to modulate the filtration of concerned nephron.[47,48]

KIDNEY INTERSTITIUM

The interstitium of the kidney is comparatively sparse, occupying 5% to 7% of the volume of the renal cortex, with a tendency to increase with age. The interstitium increases across the medulla from cortex to papilla. In the outer stripe, it is 3% to 4%, the lowest value of all kidney zones; this is interpreted as forming a barrier to prevent loss of solutes from a hyperosmolar medulla into the cortex. The interstitium is 10% in the inner stripe and up to about 30% in the inner medulla. The cellular constituents of the interstitium include resident fibroblasts (roughly 50%), which establish the scaffold frame for renal corpuscles, tubules, and blood vessels. Furthermore, dendritic cells (also roughly 50%) and varying numbers of macrophages are present. The space between the cells is filled with extracellular matrix, namely, ground substance (proteoglycans, glycoproteins), fibrils, and interstitial fluid.[49]

Fibroblasts are interconnected by specialized contacts and adhere by specific attachments to the basement membranes surrounding the tubules, renal corpuscles, capillaries, and lymphatics. They are the key cells in driving tubulointerstitial fibrosis.[50]

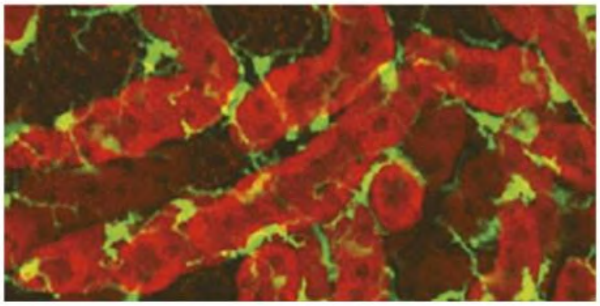

Fig. 1.15 Renal Dendritic Cells. Dendritic cells (CX_3CR1^+ cells, *green*) surrounding tubular segments in the medulla of mice (three-dimensional reconstruction). (Modified from Soos TJ, Sims TN, Barisoni L, et al. CX3CR1+ interstitial dendritic cells form a contiguous network throughout the entire kidney. *Kidney Int.* 2006;70:591–596.)

Kidney fibroblasts are difficult to distinguish from interstitial dendritic cells on a morphologic basis because both may show a stellate cellular shape and both display substantial amounts of mitochondria and endoplasmic reticulum. However, kidney fibroblasts may easily be distinguished by immunocytochemical techniques and by their subplasmalemmal actin cytoskeleton. Dendritic cells constitutively express the major histocompatibility complex class II antigen and may express antigens such as CD11c. Dendritic cells may have an important role in maintaining peripheral tolerance in the kidney (Fig. 1.15).[51] In contrast, fibroblasts in the kidney cortex (not in the medulla) contain the enzyme ecto-5′-nucleotidase (5′-NT). A subset of 5′-NT–positive fibroblasts of the cortex synthesizes erythropoietin.[52] Under normal conditions, these fibroblasts are exclusively found within the juxtamedullary portions of the cortical labyrinth. When there is an increasing demand for erythropoietin, synthesizing cells are recruited from fibroblasts in superficial portions of the cortical labyrinth.[53]

Fibroblasts within the medulla, especially within the inner medulla, have a particular phenotype known as *lipid-laden interstitial cells.* The cells are oriented strictly perpendicularly toward the longitudinal axis of the tubules and vessels (running all in parallel) and contain conspicuous lipid droplets. These fibroblasts of the inner medulla produce large amounts of glycosaminoglycans and, possibly related to the lipid droplets, vasoactive lipids, especially PGE_2.[51]

SELF-ASSESSMENT QUESTIONS

1. Which of the statements concerning the interstitium of the healthy kidney is *incorrect?*
   The renal interstitium contains:
   A. fibroblasts
   B. myofibroblasts
   C. dendritic cells
   D. cells that produce erythropoietin
   E. macrophages
2. Which of the statements concerning the podocyte foot processes is *incorrect?*
   Podocyte foot processes:
   A. contain an actin-based cytoskeleton
   B. have contractile properties
   C. are connected to the glomerular basement membrane (GBM) by integrins
   D. are connected to the GBM by the dystroglycans
   E. are connected to each other by the slit diaphragm
3. Which of the statements concerning the inner stripe of the renal medulla is *incorrect?*
   The inner stripe of the outer medulla contains:
   A. the thin descending limbs of short loops
   B. the thin descending limbs of long loops
   C. the thin ascending limbs of long loops
   D. the thick ascending loops of short loops
   E. venous vasa recta that originate in the inner medulla

REFERENCES

1. Kriz W, Kaissling B. Structural organization of the mammalian kidney. In: Seldin DW, Giebisch G, Philadelphia, eds. *The Kidney*. 3rd ed. Lippincott Williams and Wilkens; 2000:587–654.
2. Kriz W, Bankir L. A standard nomenclature for structure of the kidney. The renal commission of the International Union of Physiological Sciences (IUPS). *Pflügers Arch*. 1988;411:113–120.
3. Rollhäuser H, Kriz W, Heinke W. Das Gefässsystem der Rattenniere. *Z Zellforsch Mikrosk Anat*. 1964;64:381–403.
4. Barajas L. Innervation of the renal cortex. *Fed Proc*. 1978;37:1192–1201.
5. DiBona GF, Kopp UC. Neural control of renal function. *Physiol Rev*. 1997;77:75–197.
6. Elger M, Sakai T, Kriz W. The vascular pole of the renal glomerulus of rat. *Adv Anat Embryol Cell Biol*. 1998;139:1–98.
7. Inoue S. Ultrastructural architecture of basement membranes. *Contrib Nephrol*. 1994;107:21–28.
8. Miner JH. Renal basement membrane components. *Kidney Int*. 1999;56:2016–2024.
9. Abrahamson DR. Role of the podocyte (and glomerular epithelium) in building the GBM. *Semin Nephrol*. 2012;32:342–349.
10. Miner JH. Glomerular basement membrane composition and the filtration barrier. *Pediatr Nephrol*. 2011;26:1413–1417.
11. Kriz W, Elger M, Lemley KV, Sakai T. Structure of the glomerular mesangium: a biomechanical interpretation. *Kidney Int*. 1990;38:S2–S9.
12. Zimmerman SE, Hiremath C, Tsunezumi J, Yang Z, Finney B, Marciano DK. Nephronectin regulates mesangial cell adhesion and behavior in glomeruli. *J Am Soc Nephrol*. 2018;29. epub ahead of print.
13. Schlöndorff D, Banas B. The mesangial cell revisited: no cell is an island. *J Am Soc Nephrol*. 2009;20:1179–1187.
14. Haraldsson B, Jeansson M. Glomerular filtration barrier. *Curr Opin Nephrol Hypertens*. 2009;18:331–335.
15. Obeidat M, Obeidat M, Ballermann BJ. Glomerular endothelium: a porous sieve and formidable barrier. *Exp Cell Res*. 2012;318:964–972.
16. Neal CR, Crook H, Bell E, Harper SJ, Bates DO. Three-dimensional reconstruction of glomeruli by electron microscopy reveals a distinct restrictive urinary subpodocyte space. *J Am Soc Nephrol*. 2005;16:1223–1235.
17. Ichimura K, Miyazaki N, Sadayama S, et al. Three-dimensional architecture of podocytes revealed by block-face scanning electron microscopy. *Sci Rep*. 2015;5:8993.
18. Faul C. The podocyte cytoskeleton: key to a functioning glomerulus in health and disease. *Contrib Nephrol*. 2014;183:22–53.
19. Suleiman HY, Roth R, Jain S, Heuser JE, Shaw AS, Miner JH. Injury-induced actin cytoskeleton reorganization in podocytes revealed by super-resolution microscopy. *JCI Insight*. 2017;2.
20. Kriz W, Lemley KV. Mechanical challenges to the glomerular filtration barrier: adaptations and pathway to sclerosis. *Pediatr Nephrol*. 2017;32:405–417.
21. Pavenstädt H, Kriz W, Kretzler M. Cell biology of the glomerular podocyte. *Physiol Rev*. 2003;83:253–307.
22. Reiser J, Polu KR, Moller CC, et al. TRPC6 is a glomerular slit diaphragm-associated channel required for normal renal function. *Nat Genet*. 2005;37:739–744.
23. Winn MP, Conlon PJ, Lynn KL, et al. A mutation in the TRPC6 cation channel causes familial focal segmental glomerulosclerosis. *Science*. 2005;308:1801–1804.
24. Greka A, Mundel P. Calcium regultes podocyte actin dynamics. *Semin Nephrol*. 2012;32:319–326.
25. Kerjaschki D, Farquhar MG. Immunocytochemical localization of the Heymann antigen (gp 330) in glomerular epithelial cells of normal Lewis rats. *J Exp Med*. 1983;157:667–686.
26. Bartlett CS, Jeansson M, Quaggin SE. Vascular growth factors and glomerular disease. *Annu Rev Physiol*. 2016;78:437–461.
27. Kriz W, Lemley KV. A potential role for mechanical forces in the detachment of podocytes and the progression of CKD. *J Am Soc Nephrol*. 2015;26:258–269.
28. Endlich N, Endlich KH. The challenge and response of podocytes to glomerular hypertension. *Semin Nephrol*. 2012;32:327–341.
29. Tojo A, Hatakeyama S, Kinugasa S, Fukuda S, Sakai T. Enhanced podocyte vesicle transport in the nephrotic rat. *Med Mol Morphol*. 2017;50:86–93.
30. Hausmann R, Kuppe C, Egger H, et al. Electrical forces determine glomerular permeability. *J Am Soc Nephrol*. 2010;21:2053–2058.
31. Moeller MJ, Tenten V. Renal albumin filtration: aternative models to the standard physical barriers. *Nat Rev Nephrol*. 2013;9:266–277.
32. Kriz W, Hähnel B, Hosser H, Rösener S, Waldherr R. Structural analysis of how podocytes detach from the glomerular basement membrane under hypertrophic stress. *Front Endocrinol*. 2014;5:207.
33. Kriz W, Kaissling B. Structural organization of the mammalian kidney. In: Alpern RJ, Caplan MJ, Moe OW, eds. *Seldin and Giebisch's the Kidney: Physiology and Pathophysiology*. 5th ed. Philadelphia: Elsevier; 2013:595–691.
34. Appel D, Kershaw DB, Smeets B, et al. Recruitment of podocytes from glomerular parietal epithelial cells. *J Am Soc Nephrol*. 2009;20:333–343.
35. Peired AJ, Melica ME, Molli A, Nardi C, Romagnani P, Lasagni L. Molecular mechanisms of renal progenitor regulation: how many pieces in the puzzle? *Cells*. 2021;10.
36. Yu AS. Claudins and the kidney. *J Am Soc Nephrol*. 2015;26:11–19.
37. Maunsbach AB. Functional ultrastructure of the proximal tubule. In: Windhager EE, ed. *Handbook of Physiology: Section on Renal Physiology*. 2nd ed. New York: Oxford University Press; 1992:41–108.
38. Wade JB, Lee AJ, Liu C, et al. UT-A2: a 55-kDa urea transporter in thin descending limb whose abundance is regulated by vasopressin. *Am J Physiol Renal Physiol*. 2000;278:F52–F62.
39. Marcoux AA, Tremblay LE, Slimani S, et al. Molecular characteristics and physiological roles of Na(+) -K(+) -Cl(-) cotransporter 2. *J Cell Physiol*. 2021;236:1712–1729.
40. Nie M, Bal MS, Liu J, et al. Uromodulin regulates renal magnesium homeostasis through the ion channel transient receptor potential melastatin 6 (TRPM6). *J Biol Chem*. 2018;293:16488–16502.
41. Sabolic I, Brown D. Water channels in renal and nonrenal tissues. *News Physiol Sci*. 1995;10:12–17.
42. Bankir L, Trinh-Trang-Tan MM. Urea and the kidney. In: Brenner BM, ed. *The Kidney*. 6th ed. Philadelphia: Saunders; 2000:637–679.
43. Geng X, Zhang S, He J, et al. The urea transporter UT-A1 plays a predominant role in a urea-dependent urine-concentrating mechanism. *J Biol Chem*. 2020;295:9893–9900.
44. Madsen KM, Verlander JW, Kim J, Tisher CC. Morphological adaptation of the collecting duct to acid-base disturbances. *Kidney Int*. 1991;40:S57–S63.
45. Mundel P, Bachmann S, Bader M, et al. Expression of nitric oxide synthase in kidney macula densa cells. *Kidney Int*. 1992;42:1017–1019.
46. Harris RC, McKanna JA, Akai Y, et al. Cyclooxygenase-2 is associated with the macula densa of rat kidney and increases with salt restriction. *J Clin Invest*. 1994;94:2504–2510.
47. Kurtz A. Renin release: sites, mechanisms, and control. *Annu Rev Physiol*. 2011;73:377–399.
48. Schnermann JB, Briggs JD. Tubular control of renin synthesis and secretion. *Pflugers Arch*. 2013;465:39–51.
49. Kaissling B, Hegyi I, Loffing J, Le Hir M. Morphology of interstitial cells in the healthy kidney. *Anat Embryol*. 1996;193:303–318.
50. Kuppe C, Ibrahim MM, Kranz J, et al. Decoding myofibroblast origins in human kidney fibrosis. *Nature*. 2021;589(7841):281–286.
51. Krüger T, Benke D, Eitner F, et al. Identification and functional characterization of dendritic cells in the healthy murine kidney and in experimental glomerulonephritis. *J Am Soc Nephrol*. 2004;15:613–621.
52. Bachmann S, Le Hir M, Eckardt KU. Co-localization of erythropoietin mRNA and ecto-5'-nucleotidase immunoreactivity in peritubular cells of rat renal cortex indicates that fibroblasts produce erythropoietin. *J Histochem Cytochem*. 1993;41:335–341.
53. Kaissling B, Spiess S, Rinne B, Le Hir M. Effects of anemia on the morphology of the renal cortex of rats. *Am J Physiol*. 1993;264:F608–F617.